

Diego Royo Meneses

PhD Candidate @ Graphics and Imaging Lab, Universidad de Zaragoza, Spain

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🎓 Google Scholar: [Diego Royo](#)

Hi! :-) I'm Diego, a PhD candidate in the Graphics and Imaging Lab, under the supervision of [Prof. Adolfo Muñoz](#) and [Prof. Diego Gutiérrez](#). I previously got my BSc and MEng in Computer Science at Universidad de Zaragoza.

My research interests span light transport, computational imaging and astronomy. I really enjoy creating projects driven by curiosity, delving into new topics, and ensuring that they are open, free and usable by other people.

I have great experience with Python (including Pytorch) and C++ (with my own autodiff engine). [Check out my projects!](#)

education

Doctoral Program in Systems and Informatics Engineering

jan. 2022 - jan. 2026

EINA, Universidad de Zaragoza

Zaragoza, Spain

Research interests: Light transport, computational imaging and astronomy.

Supervised by [Prof. Adolfo Muñoz](#) and [Prof. Diego Gutiérrez](#).

Funded by a 4-year competitive grant (Gobierno de Aragón).

Master Program in Robotics, Graphics and Computer Vision

sep. 2020 - feb. 2022

EINA, Universidad de Zaragoza

Zaragoza, Spain

Maximum grade with honors. GPA: 9,86/10.

📌 Thesis: [Optical flow estimation in systems with co-located light and camera for monocular endoscopy](#).

Degree in Informatics Engineering

sep. 2016 - jul. 2020

EINA, Universidad de Zaragoza

Zaragoza, Spain

Maximum grade with honors. GPA: 9,34/10.

📌 Thesis: [Numerical estimation of differential properties of the deformation field in deformable SLAM with FlowNet2](#).

publications

journal

(J1) Virtual Mirrors: Non-line-of-sight imaging beyond the third bounce

[Diego Royo](#), Talha Sultan, Adolfo Muñoz, Khadijeh Masumnia-Bisheh, Eric Brandt, Diego Gutierrez, Andreas Velten, Julio Marco

[JCR Q1] Transactions on Graphics (Proceedings of SIGGRAPH), 2023

DOI: <https://doi.org/10.1145/3592429>

(J2) Structure-aware parametric representations for time-resolved light transport

[Diego Royo](#), Zesheng Huang, Yun Liang, Boyan Song, Adolfo Muñoz, Diego Gutierrez, Julio Marco

[JCR Q1] Optics Letters, 2022, Vol. 47, Iss. 19

DOI: <https://doi.org/10.1364/OL.465316>

(J3) Non-line-of-sight transient rendering

[Diego Royo](#), Jorge García-Pueyo, Adolfo Muñoz, Adrián Jarabo

★ Best Full Paper Award (CEIG)

[JCR Q2] Computers & Graphics (CEIG), 2022, Vol. 107

DOI: <https://doi.org/10.1016/j.cag.2022.07.003>

poster

(P1) mitransient: Transient light transport in Mitsuba 3

[Diego Royo](#), Miguel Crespo, Jorge García-Pueyo

★ Best Poster Award (ICCP 2024)

International Conference on Computational Photography (ICCP 2024)

DOI: <https://doi.org/10.5281/zenodo.11032518>

funding and awards

NASA Space Apps 2019 - World finalists

oct. 2019

Member of *Need Another Seven Astronauts*, winner team in Zaragoza and one of thirty-six world finalists chosen by NASA. Developed a satellite tracking app which made it to local newspapers.

 [Click here to see our project and media coverage](#)

NASA Space Apps 2022 - World finalists

oct. 2022

Member of *Sin Rocket no hay Paraíso*, winner team in Zaragoza and one of thirty-five world finalists chosen by NASA (again! and this time with a sillier name). Developed a fire spread prediction algorithm which made it to national/local newspapers and radio.

 [Click here to see our project and media coverage](#)

Outstanding Achievement Award (MEng)

sep. 2021

Best academic record during my Master studies.

DGA Fellowship for Predoctoral Studies

dec. 2021

Four-year competitive grant from Gobierno de Aragón in Spain. Ranked #1 out of 55 awardees.

CEIG 2022 - Best full paper award

jun. 2022

Awarded to *Non-line-of-sight transient rendering*.
Congreso Español de Informática Gráfica (CEIG 2022), Spain.

JJI 2023 - Best contribution (ICT)

jun. 2023

Awarded to *Virtual Mirrors: Non-line-of-sight imaging beyond the third bounce*.
Jornada de Jóvenes Investigadores (JJI 2023), Spain.

ICCP 2024 - Best poster award

jul. 2024

Awarded to *mitransient: Transient path tracing in Mitsuba 3*.
International Conference on Computational Photography (ICCP 2024), Lausanne, Switzerland.

research experience

Computational Cameras Group

sep. 2024 - dec. 2024

California Institute of Technology

Pasadena, CA, United States

Three-month visit under the supervision of [Prof. Katherine L. Bouman](#).

Research on advanced computational imaging techniques that enable mass mapping of galaxy clusters using weak and strong gravitational lensing.

Graphics and Imaging Lab

sep. 2021 - now

EINA, Universidad de Zaragoza

Zaragoza, Spain

During my thesis, I've worked (and keep working) on transient light transport simulation, and computational imaging methods that enable looking around corners.

 *mitransient*: Transient light transport simulator ★ Starred 35 times

 [Project website & documentation](#)

 *mitsuba2-transient-nlos*: Older version of *mitransient* ★ Starred 24 times

 *tal*: Toolkit for simulation and analysis of non-line-of-sight captures ★ Starred 20 times

EndoMapper

oct. 2019 - feb. 2022

EINA, Universidad de Zaragoza

Zaragoza, Spain

Research tasks in collaboration with the Department of Computer Science and Systems Engineering (DIIS) on the EndoMapper project. Focused on optical flow estimation in monocular endoscopy videos.

 *flowvid*: Toolkit for visualizing and operating with optical flow ★ Starred 30 times

So far, I've received the maximum evaluation score from the students in every academic course.

2024-25

Computational Imaging

Master in Robotics, Graphics and Computer Vision
EINA, Universidad de Zaragoza, Spain

2024-25

Advanced Computational Imaging

Master in Robotics, Graphics and Computer Vision
EINA, Universidad de Zaragoza, Spain

2022-23 & 2023-24

Programming Fundamentals

Bachelor's Degree in Industrial Technologies
EINA, Universidad de Zaragoza, Spain

2022-23 & 2023-24

Computer Graphics

Bachelor's Degree in Informatics Engineering

-  **Render contest for students - courses 22-23 & 23-24**
EINA, Universidad de Zaragoza, Spain

2023-24

Modeling and Simulation of Appearance

Master in Robotics, Graphics and Computer Vision

-  **Render contest for students - course 23-24**
EINA, Universidad de Zaragoza, Spain

 dissemination

2022 & 2023

European Researchers' Night

Dissemination of the Graphics & Imaging Lab's research areas with interactive experiments targeted at children and families.

2022 & 2023 & 2024

Semana de la Ingeniería y Arquitectura, Girls' Day

High school students come to the Graphics & Imaging Lab to learn about science and research.

 supervision

undergrad theses

Sampling techniques for non-line-of-sight light transport simulation

Ismael Tienda (jun. 2023 - sep. 2023)
BSc Computer Science. **Grade: 9.5/10**

Simulation of ultra-fast photon capture hardware in light transport

Jorge Solán (nov. 2023 - sep. 2024)
BSc Computer Science. **Grade: 9/10**

other

Dark Matter Diffusion

Supervision for CS 163 - Projects in Machine Learning
Computing and Mathematical Sciences
California Institute of Technology, United States

internships

Real-time NLOS reconstruction using hardware acceleration

Andrey Pushkar (feb. 2024 - may. 2024)
MEng Robotics, Graphics and Computer Vision

Performance analysis of Virtual Mirror surfaces

Salvador Rodríguez (oct. 2023 - jan. 2024)
BSc Mathematics

Signal compression for non-line-of-sight imaging

Ayush Gupta (may. 2023 - jul. 2023)
BSc Computer Science at IIT Delhi, India

 personal projects

I like tinkering and building things that can be useful for other people, or just for fun.

Check out [my personal website](#) for some examples.

This CV template was made by me from scratch :-)

 languages

Spanish Native language

English C1 Advanced Certificate. Cambridge.
Grade A with CEFR C2 level.

I'm also currently learning Japanese.